



$$W = 6$$

$$val = [4, 3, 6, 2], \quad wt = [2, 1, 4, 3]$$

$$(2, 1, 3) \Rightarrow 4 + 3 + 2 = 9$$

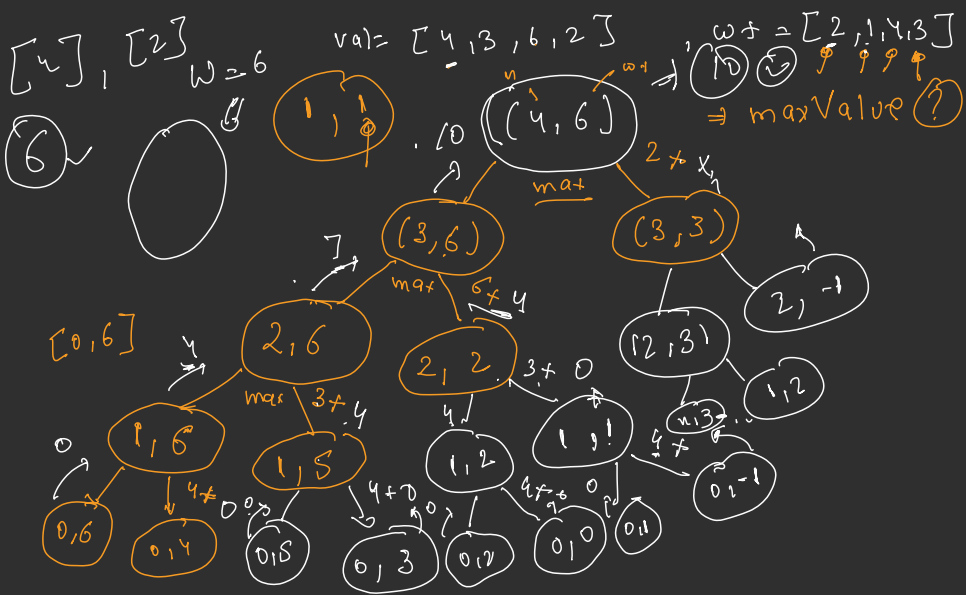
$$(2, 4) \Rightarrow 4 + 6 = 10$$

$$(2, 3) \Rightarrow 5 \quad 4 + 2 = 6$$

$$(1, 4) \Rightarrow 5 \quad 3 + 6 = 9$$

ans: 6
✓
10

$$[2] \quad [4] \quad [4, 3] \quad [2, 1]$$



```

int findMaxVal (int n, int w, wt, val)
{
    if (w < 0)
        return INT_MIN;
    if (n == 0 || w == 0)
        return 0;

```

```

    return max ( findMaxVal (n-1, w, wt, val),
                val[n-1] + findMaxVal (n-1, w-wt[n-1], wt, val) );

```

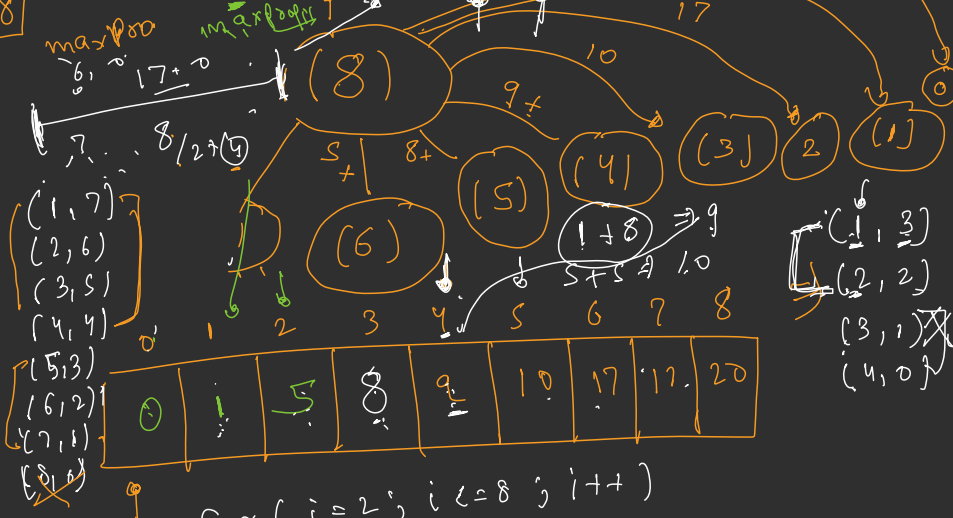
0	0	0	0	0	0	0	0
1	0						
2	0						
3	0						
4	0						

$$dp[i][j] = \max \left[dp[i-1][j], val[i-1] + dp[i-1][j - w[i-1]] \right]$$

len = 8

Price: [1, 5, 8, 9, 10, 17, 17, 20]

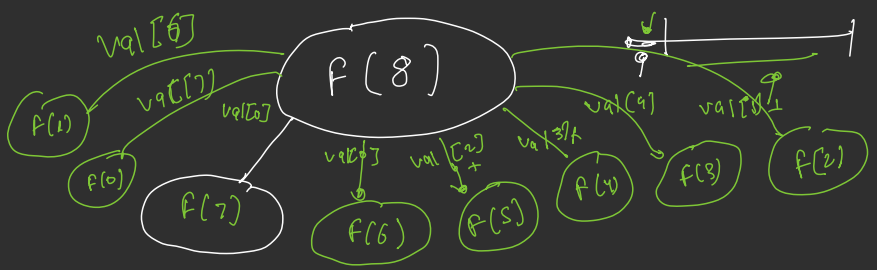
(1)



```

for (i = 2; i <= 8; i++)
{
    for (int j = 1; j <= i/2; j++)
        dp[i] = max(dp[i], dp[j] + dp[i-j])
}

```



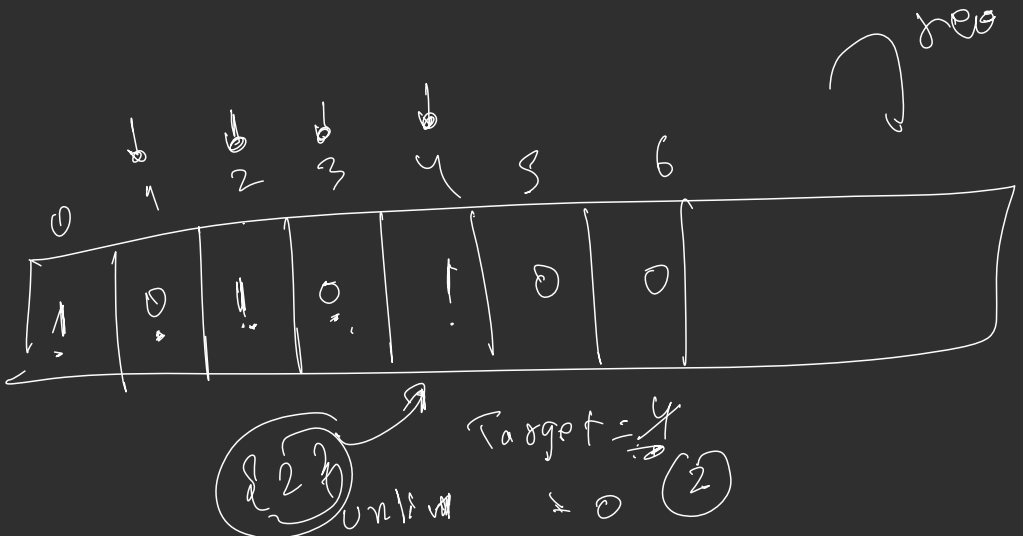
[31, 24, 33, 21, 40]

4 second

Target

15

2
75



{2, 3, 1}

{2}

{2, 3, 1, 4, 5, 6}

(2) (3)

(2) $\rightarrow$ (4-2) $\rightarrow$ 2

{3} 8

(1)

{2, 1}

max
(1) (2)

1	1	1	1	1	1	1	0	0	1	0
---	---	---	---	---	---	---	---	---	---	---

1D array
(2) - 12 13

como n
sum = target